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PALEOMAGNETISM AND CRUSTAL ROTATIONS ALONG A SHEAR ZONE, LAS VEGAS RANGE, SOUTHERN NEVADA

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Abstract. In southern Nevada, Paleozoic isopachs, Mesozoic structures, and mountain ranges trend north-south, but as they approach the right-lateral, west-northwest-trending Las Vegas Valley Shear Zone (LVVSZ), they are bent clockwise up to 90° in oroflexes 30 km long. In the Las Vegas Range the declination of the characteristic magnetization of the Cambrian Bonanza King Formation has likewise been rotated clockwise more than 50°. For samples from eight sites (13-23 samples per site) in the Las Vegas Range and in its extension south of the LVVSZ, stepwise thermal demagnetization to 600°C removed recent and Tertiary overprints, revealing a nearly horizontal, late Paleozoic viscous partial thermoremanent magnetization (VPTRM) which predates both the Tertiary oroflexural bending and the Sevier thrusting in the area. The declination of the characteristic magnetization (after correcting for the dip of bedding) varies from 133° ± 2° at a site 24 km north of the shear zone to 196° ± 4° 13 km north of it and 204° ± 4° 11 km south of the shear zone. This parallels a similar change in the structural trend. The paleomagnetic data indicate that the Las Vegas Range is broken into rather small blocks (<5 km) which rotate in response to more continuous deformation at depth. This deformation of the upper crust can be modelled using a thin non-Newtonian viscous plate model, which when applied to the Las Vegas Valley Shear Zone predicts rotations similar to those seen in the Las Vegas Range.

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INTRODUCTION

The Las Vegas Valley Shear Zone (LVVSZ) is a major right-lateral fault zone trending northwest from Las Vegas, Nevada. Movement along the fault has offset the north-south trending mountain ranges of southern Nevada (Figure 1) and bent them into large "oroflexes," which superficially resemble giant drag folds. The oroflex at the southern end of the Las Vegas Range north of the shear zone is particularly striking. Within 30 km the range bends almost 90° clockwise, its trend changing from north-south to roughly east-west near the fault. This area provides an excellent setting for an investigation of oroflexural bending and large-scale deformation in general. Previous paleomagnetic work has shown that the rocks of the Cambrian Bonanza King Formation acquired a stable remanent magnetization in late Paleozoic time, long before the deformation associated with the Miocene movement along the LVVSZ

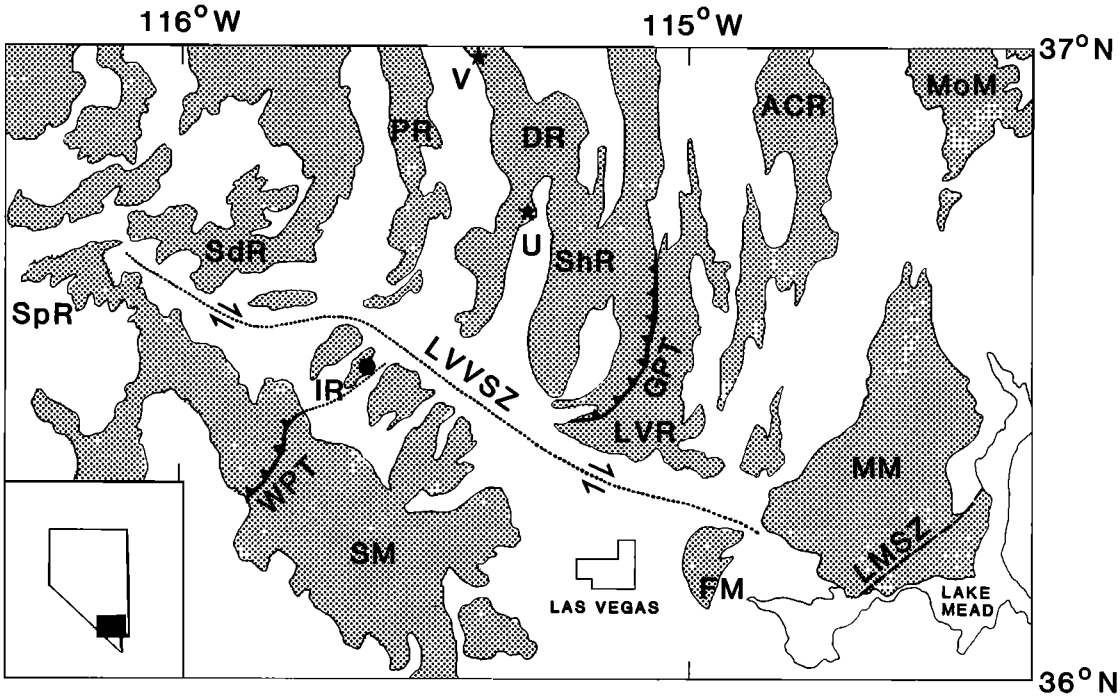


Fig. 1. Index map of southern Nevada. Mountain ranges shown are: ACR, Arrow Canyon Range; DR, Desert Range; FM, Frenchman Mountain; LVR, Las Vegas Range; MM, Muddy Mountains; MoM, Mormon Mountains; PR, Pintwater Range; SM, Spring Mountains; ShR, Sheep Range; SdR, Spotted Range; and SpR, Specter Range. Faults indicated are the Wheeler Pass Thrust (WPT), Gass Peak Thrust (GPT), Las Vegas Valley Shear Zone (LVVSZ), and the Lake Mead Shear Zone (LMSZ). Solid circle marks sample site at Indian Ridge (IR). Stars mark sample sites U and V of Gillett and Van Alstine [1982].

[Gillett and Van Alstine, 1979, 1982]. The direction of this magnetization can thus be used as a marker to determine how the Las Vegas Range has been deformed.

In this study we not only document structural rotation but also use paleomagnetism to try to understand the extent and mechanism of that rotation. Researchers frequently report rotated paleomagnetic declinations from a few sites and conclude that terranes several tens or hundreds of kilometers in extent have been similarly rotated. In areas where the geologic structure is well exposed and well mapped and where more thorough paleomagnetic sampling has been done, rotations due to local structures (≈10 km in size) have been documented [e.g. Channell et al., 1978; Grubbs and Van der Voo, 1976; Reidel et al., 1984]. Studies of rotations in areas where the geologic structure is well-known, like the Las Vegas Range, are essential if we are to evaluate estimates of rotations in poorly exposed, less well-understood areas. In addition, the data presented here provide a test of recently published models for deformation near shear zones [Ron et al., 1984; McKenzie and Jackson, 1983; England et al., 1985].

GEOLOGIC SETTING

Structural trends, facies boundaries, and Paleozoic isopachs are all bent and offset right-laterally across the LVVSZ [Stewart et al., 1968]. Longwell [1974] showed that the shear zone offsets the Miocene (≈17 Ma) Thumb Formation right-laterally more than 64 km. Ekren et al. [1968] concluded from their work on the Nevada Test Site that much of the movement on the LVVSZ has occurred since 17 Ma. Movement must have ended by early Pliocene time when deposition of the Muddy Creek Formation began, since it is not cut by the shear zone [Longwell, 1974].

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Burchfiel [1965] correlated the Gass Peak Thrust in the Las Vegas Range with the Wheeler Pass Thrust, east of Indian Ridge, south of the LVVSZ (Figure 1). Both thrusts put Precambrian Stirling Quartzite on the late Paleozoic Bird Spring Formation. The shear zone offsets the thrust 22 km, and bending north of the shear zone accounts for 21 km of additional displacement, for a minimum of 43 km of offset. In contrast, about 80 km to the west of the Las Vegas Range, in the Specter Range, Burchfiel found that movement along the shear zone was taken up entirely by bending, not by faulting. The shear zone does not cut the Specter Range at the surface and is not continuous with the Walker Lane to the northwest.

The LVVSZ is thought to be an intracontinental transform separating regions of differential extension [Davis and Burchfiel, 1973; Anderson, 1973; Wernicke et al., 1982]. Detailed mapping by Guth [1981] in the Sheep, Desert, and Las Vegas ranges indicates that the region north of the shear zone and west of the Las Vegas Range has undergone almost 100% extension in Miocene time. The Spring Mountains, south of the shear zone, have not undergone any significant extension [Burchfiel et al., 1974].

Landsat images of southern Nevada show that mountain ranges along the Las Vegas Valley have been bent into "oroflexes," arcuate ranges suggesting giant drag folds [Albers, 1967; Burchfiel, 1965] caused by right-lateral movement across the LVVSZ. The most striking oroflex along the LVVSZ occurs at the southern end of the Las Vegas Range (Figure 1). At the southernmost end of the range, 30 km north of the city of Las Vegas, the range trends roughly east-west. Farther away from the shear zone, structures trend more northeasterly, and by 30 km north from the shear zone the range runs north-south, following the regional trend of nearby ranges. The structural geology and stratigraphy of the Las Vegas Range have been described in detail by Guth [1980] and Ebanks [1965].

The Las Vegas Range consists of a 6-km-thick section of uppermost Precambrian to Permian miogeosynclinal sedimentary rocks, which are exposed continuously along the range to within 5 km of the LVVSZ. They include thick sections of shallow marine limestones such as the Cambrian Bonanza King, the Cambrian Nopah, and the Permo-Pennsylvanian Bird Spring Formations. Higher up in the section is a regional unconformity due to widespread uplift in late Permian time. The Mesozoic section consists mostly of continental red beds and sandstones.

During the Mesozoic Sevier orogeny, the Gass Peak Thrust and several other eastward-directed thrusts (Figure 1) became active. The Gass Peak Thrust runs along the axis of the Las Vegas Range and places uppermost Precambrian Stirling Quartzite on the Bird Spring Formation. Guth [1980] documented 5900 m of stratigraphic displacement on the Gass Peak Thrust and thought that horizontal displacement on it exceeded 30 km. A number of folds with horizontal axes parallel to the Gass Peak Thrust formed during thrusting, especially in the Bird Spring Formation of the lower plate.

During Neogene time, extension of the Basin and Range was accompanied by listric normal faulting, creating the north-south mountain ranges that we see today. The Mormon Pass Fault, a major, west dipping normal fault, runs along the western edge of the Las Vegas Range, separating the Las Vegas Range, which was relatively unaffected by Basin and Range faulting, from the Desert and Sheep ranges which have undergone ≈100% east-west extension [Guth, 1980, 1981]. Alluvial conglomerates, lacustrine limestones, and rhyolitic tuffs of the Miocene Horse Spring Formation were deposited in the valleys between the ranges. During Miocene time, movement of the LVVSZ offset and deformed both the Tertiary normal faults and earlier Sevier age thrusts and folds. Basin and Range extension continued both during and after movement along the LVVSZ.

Gillett and Van Alstine [1979, 1982] and Van Alstine and Gillett [1979] studied the latest Precambrian and earliest Paleozoic strata of the Desert Range, 25 km west of the Las Vegas Range (Figure 1). They found that the rocks of the Upper Cambrian Bonanza King Formation have a characteristic magnetization (southeasterly declination, I ≈ -20°) apparently due to remagnetization during the late Paleozoic. A less stable Tertiary or late Cretaceous component was removed using thermal demagnetization. The declination of the characteristic component varied between sites, suggesting that the rocks of the northern end of the Desert Range have been rotated

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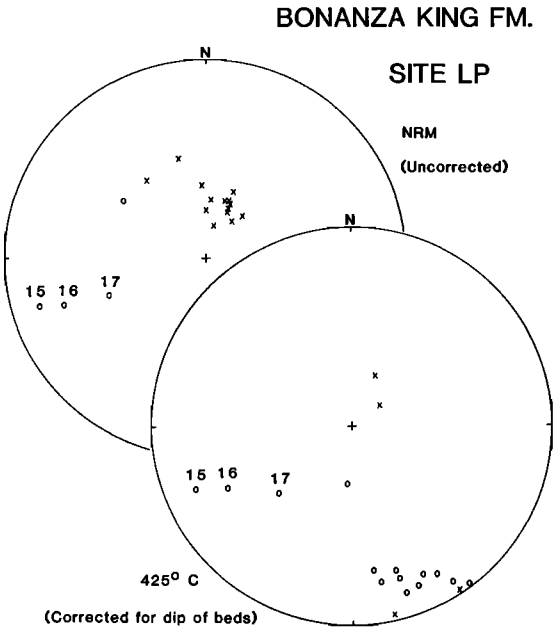


Fig. 2. Equal area plots of the magnetization directions for samples from site LP, before and after demagnetization to 425°C. Open circles plot on the upper hemisphere, x plot on the lower hemisphere. Samples 15-17 have been remagnetized by lightning strikes. Only data from the 12 samples plotting in the south eastern quadrant after demagnetization were used to determine the mean direction of the characteristic component.

44° ± 5° counterclockwise with respect to those 30 km to the south.

PRESENT STUDY

Methodology

The Cambrian Bonanza King Formation in the Las Vegas Range consists of 900 m [Guth, 1980] of shallow marine limestones and dolomites and is exposed along most of the length of the Las Vegas Range. Dark gray lime mudstone, often mottled due to bioturbation, is the predominant rock type. Siltier, limonite-stained limestone commonly occurs as stringers or wavy laminae in the dark limestone. The rock commonly contains stylolites and calcite-filled fractures, which were not sampled when possible. Most samples were taken from the dark grey limestone, which also gave the most consistent paleomagnetic results.

Samples were taken from sixteen sites in the Las Vegas Range. In addition, samples were collected at one site at Indian Ridge (IR), 25 km west of the Las Vegas Range, about 10 km south of the LVVSZ (Figure 1). Care was taken to sample at well-exposed sites away from major faults and folds associated with Sevier age thrusting and other structural complications. Between 13 and 23 samples were collected at each site. Samples were usually spaced 1-3 m apart and typically spanned 10-20 m of stratigraphic section.

At most sites, samples were collected using a gasoline-powered core drill. Cores were oriented with a Brunton compass. At six sites (IR, PC, MR, QS, BL, and GV), block samples were collected using the technique described by Gillett and Van Alstine [1979]. Cores were drilled from the block samples using a drill press in the lab. Each core, 2.5 cm in diameter, was cut into specimens 2 to 3 cm long, which were then rinsed in dilute HCl for 5-10 s to remove any metallic contamination from the drill bit or the rock saw used to trim the specimens. This is particularly important because these weakly magnetized rocks have natural remanent magnetization (NRM) intensities of $\approx 5 \times 10^{-5}$ Am^2/kg.

All measurements were made in the paleomagnetism lab at the California Institute of Technology using an ScT cryogenic magnetometer in a shielded room where the strength of the ambient field was ≈100 nT. All specimens were measured and thermally demagnetized inside the shielded room. Data collection and reduction were done with an Apple IIe computer interfaced to the magnetometer.

In addition to measuring the remanent magnetization of the specimens, isothermal remanent magnetization acquisition studies were done to determine the magnetic minerals present in selected samples. An impulse magnetizer was used to magnetize samples using progressively higher fields, up to 0.6 T.

Paleomagnetism of the Bonanza King Formation

For most sites the NRM directions clustered around the present axial dipole field direction (e.g., site LP, Figure 2) before correcting for the dip of bedding. Thermal demagnetization was necessary to remove magnetic overprints and to isolate

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the characteristic magnetization. Most specimens were thermally demagnetized in 10-12 steps to 600°C.

Upon demagnetization to 400°C, samples from 12 sites revealed a south or southeasterly, nearly horizontal characteristic direction (e.g., site LP, Figure 2). At most sites there were a number of samples that were strongly magnetized with only a single, randomly oriented component of magnetization, which is typical of samples that have been remagnetized by lightning. Commonly, a group of adjacent samples all appear to have been lightning-struck (for instance, samples 15-17 on Figure 2). At two of the more exposed sites (MR and MW), where an older southerly component was evident, scatter apparently due to lightning strikes made it impossible to determine the remanent direction accurately (±20°). At two other sites (EC, SM) the direction of the characteristic component could not be determined accurately because most of the samples were heavily overprinted. Even at the most protected sites, well away from ridges, typically 10-20%, and in one case 70%, of the specimens have been remagnetized, presumably by lightning strikes and weathering. Such clearly remagnetized samples were not used in calculating the mean direction of the characteristic magnetization.

At the remaining five sites (BT, PC, QS, BL, GV), thermal demagnetization failed to reveal a consistent magnetization direction. At two of these sites (QS and BL), samples were drilled from blocks of a suspected cave breccia in the lower Bonanza King Formation [Gillett, 1982]. For these sites the magnetization directions were scattered, as would be expected if the magnetization predated the brecciation [Gillett, 1982]. At the other three sites where the characteristic magnetization could not be isolated, it is probable that weathering and lightning strikes have obliterated the ancient magnetization.

Specimens from the eight sites where the effects of lightning strikes could be minimized had quite similar thermal demagnetization paths. Three components of magnetization were present: (1) a very soft, viscous component aligned with the present earth field and removed by heating to 200°C, (2) an intermediate component, which is removed by demagnetization to 350°C, oriented north or northwest with a moderate (≈60°) inclination, and (3) the characteristic magnetization left after heating to 350°C, which has a south to southeasterly declination and a shallow, negative inclination. A modified Zijderveld diagram [Roy and Park, 1974; Zijderveld, 1967] of a representative sample (Figure 3) shows these three components clearly. As the specimen was demagnetized and the north-northeast modern field component was removed, the magnetization shifted westward. As the north-northwesterly intermediate component was removed, the magnetization became more southerly and shallow until only the southerly, nearly horizontal characteristic component remained. To determine accurately the direction of these different components (Table 1), we used a least squares fit program [Kirschvink, 1980]. The intermediate component represents the magnetization removed between 200° and 350°C, and the characteristic component was removed between 350° and 500°C.

Bingham statistics [Onstott, 1980] were used to calculate the mean and 95% error ellipse for both components of magnetization at each site. At several sites the magnetization directions were not distributed according to a Fisherian distribution [Fisher, 1953], possibly because thermal demagnetization did not completely remove a present-day or Tertiary overprint in some samples. The uncertainty quoted for the declination of a component equals half the total range in declination of the Bingham error oval. As Demarest [1983] pointed out for a Fisherian distribution, this will overestimate the uncertainty in declination by ≈25%.

The characteristic magnetization of these rocks is carried in magnetite. Heating the rocks to 600°C (just above the Curie temperature of magnetite) destroys more that 95% of the natural remanent magnetization (NRM), making it impossible to measure the remaining remanent field accurately. Studies of isothermal remanent magnetization (IRM) acquisition for four representative samples showed the rapid acquisition of IRM typical of magnetite-bearing rocks. Gillett and Van Alstine [1982], on the basis of alternating-field demagnetization behavior, bulk susceptibility measurements, and petrographic evidence, concluded that magnetite is the carrier of both the

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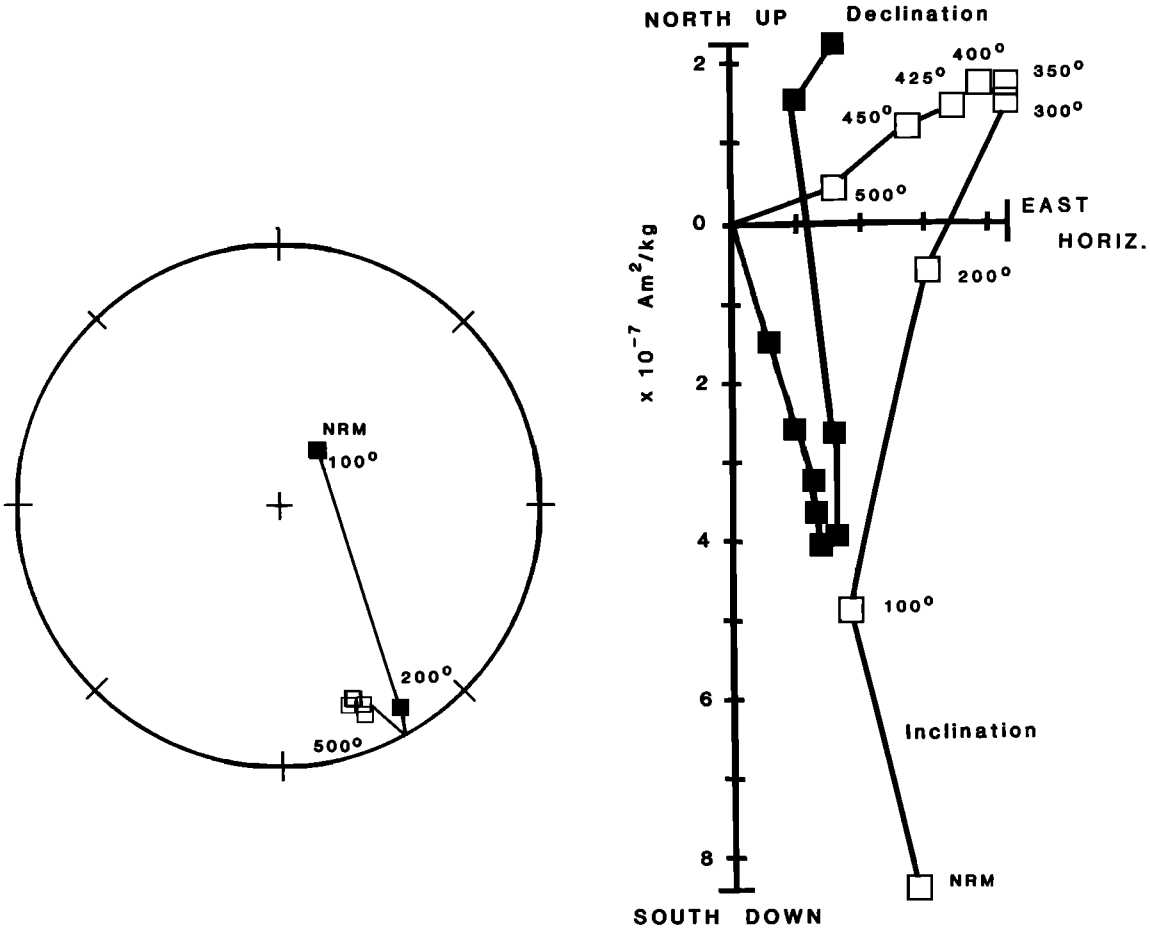


Fig. 3. Thermal demagnetization path for a representative sample from site LP on both an equal area plot and a modified Zijderveld plot using in situ coordinates. On the equal area plot the solid squares lie on the lower hemisphere, and the open squares lie on the upper hemisphere. On the modified Zijderveld plot the solid symbols represent successive positions of the end point of the magnetization vector during stepwise thermal demagnetization projected onto the horizontal (north, east) plane. The open symbols represent a plot of the horizontal and vertical components of the magnetization direction at each demagnetization step.

characteristic and the intermediate components of magnetization in the Bonanza King Formation of the Desert Range.

Paleomagnetism of Other Formations in the Las Vegas Range

Samples from the Permo-Pennsylvanian Bird Spring Formation were collected at two sites in the Las Vegas Range, EW and EP, 12 and 45 km north of the shear zone, respectively, to determine whether these limestones had an ancient magnetization that could be used as a structural marker. These samples were very weakly magnetized (NRMs with intensities of ≈5 x 10^{-8} emu/g) in the direction of the present axial dipole field. After thermal demagnetization to 400°C the specimens' magnetization was of the order of the noise level of the magnetometer (≈10^{-9} emu/g). Because of these unpromising results, no further sampling was done in the Bird Spring Formation.

The Miocene Horse Spring Formation, deposited while the LVVSZ was active, is exposed near the north and south ends of

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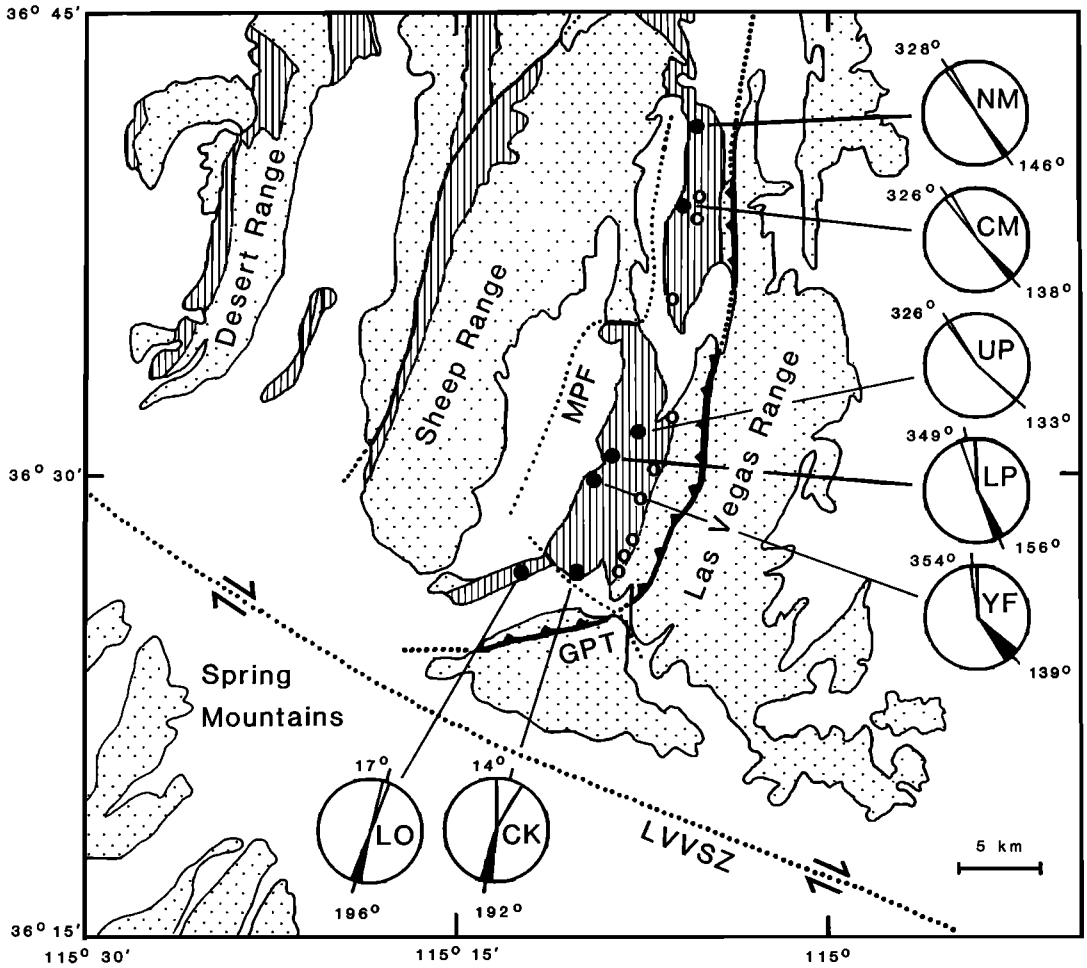


Fig. 4. Summary diagram of the declination of both the intermediate and the characteristic magnetization direction for the seven (lettered) sites in the Las Vegas Range. The angle subtended by the darkened wedge indicates the 95% uncertainty in the declination of the characteristic component, the uncolored wedge subtends an angle equal to the 95% uncertainty in the declination of the intermediate component. The stippling indicates Paleozoic bedrock outcrop. The striped region indicates outcrop of Cambrian Bonanza King Formation. The Las Vegas Valley Shear Zone (LVVSZ), the Gass Peak Thrust (GPT), and the Mormon Pass Fault (MPF) are also shown. An eighth sample site at Indian Ridge lies south of the LVVSZ, 8 km west of the west edge of the map. Open circles indicate sample sites where a consistent magnetization direction could not be determined. Geology from Guth [1980], Ebanks [1965], and Longwell et al. [1965].

the Las Vegas Range. Lacustrine limestones in the Horse Spring Formation were sampled at 3 sites, south of Fossil Ridge, near site LO, 12 km from the shear zone (Figure 4). The magnetization was aligned with the present earth field and apparently resided in goethite because it was destroyed upon heating to 100°C. There was no evidence of an ancient magnetization.

East of Mormon Pass and site CM, near the north end of the Las Vegas Range (Figure 4), rhyolitic tuff beds up to 3 m thick occur in thick deposits of fanglomerate made up of Paleozoic limestone clasts. Guth [1980] identified

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these beds as Horse Spring Formation. The tuffs were sampled at two sites 35 km north of the shear zone. Samples of a coarse-grained tuff (≈1-mm grain size) at the first site yielded randomly oriented magnetization directions and had very erratic demagnetization paths. At the second site, fifteen samples from four beds of finer-grained (≈0.3-mm grain size) reworked tuff were collected. These had NRM directions aligned with the present field. Upon demagnetization to 400°C all four samples from one of the tuff beds revealed a southerly magnetization direction with an inclination of about -60° after a tilt correction. After demagnetization at 400°C and a structural correction for the 25° dip of bedding, the magnetization directions for the remaining 11 samples grouped around a Tertiary pole of normal polarity. These tuffs probably record a primary magnetization imposed before the beds were tilted. The paleomagnetic evidence implies that these rocks have not been rotated more than 20°. Unfortunately, these tuffs are not exposed farther south.

INTERPRETATION

The characteristic magnetization of the Bonanza King Formation in the Las Vegas Range is similar to that measured by Gillett and Van Alstine [1982] for the Desert Range. At their northernmost site (V, see Figure 1), 52 km from the shear zone, the rocks of the Bonanza King Formation have a characteristic magnetization (after tilt correction) of D = 121°, I = -18°, and α_{95} = 2°, for 42 samples. At their southernmost site (U, see Figure 1), 28 km north of the shear zone, they found a characteristic magnetization of D = 165°, I = -22°, and α_{95} = 4°, for 47 samples. The characteristic magnetization measured in the Las Vegas Range (after tilt correction) varied from D = 133°, I = -10.2° for site UP to I = 196°, D = -6.8° for site LO. This magnetization was presumably acquired as a viscous partial thermoremanent magnetization when the rocks were uplifted and cooled during Late Permian time [Gillett and Van Alstine, 1982]. The Late Permian paleopole for cratonic North America is 49° N, 121° E [Irving and Irving, 1982]. Because the field was reversed throughout most of the Permian [Irving, 1966], the field direction for the Desert and Las Vegas Ranges would have been about D = 148°, I = -15.8° (Figure 5a).

At the three northernmost sites, NM, CM, and UP, both the intermediate and the characteristic magnetization directions are quite consistent (± 8°), indicating that there has been little rotation between these sites. The mean characteristic magnetization direction (D = 139°, I = -10°, after tilt correction) is very close to the Permian field direction for cratonic North America (Figure 5a). The slightly shallower inclination may be explained if the strata were tilted gently (<10°) to the west before they were remagnetized in the late Paleozoic. The regional stratigraphy indicates that this was likely [Gillett and Van Alstine, 1982]. If all of the shallow present-day dip (5°-10° to the west) was acquired by the late Paleozoic, then the mean in situ direction (D = 140°, I = -15.8°) would be a more accurate measure of the field direction when the rocks were remagnetized. Regardless, correcting for this minor amount of regional tilting does not significantly change the declination of the characteristic magnetization. An equal angle plot of the field direction for the Las Vegas Range since the late Paleozoic (Figure 5a) shows clearly that this magnetization must have been acquired before early Jurassic time, after which the field inclination was much steeper than that of the characteristic magnetization (with or without correcting for the dip of bedding).

At the remaining sites the inclination of the characteristic magnetization varies with the dip of the beds, indicating that the magnetization was acquired before the beds were tilted to their present attitude. If the acquisition of magnetization preceded tilting, then the variance of the mean inclination for all eight sites should decrease after the tilt correction is made. Indeed, the characteristic component passes this inclination-only fold test at the 99% confidence level (Table 2) [McFadden and Reid, 1982; McElhinny, 1964].

The changes in the declination of the characteristic component between sites in the Las Vegas Range are primarily due to rotations related to motion on the Las Vegas Valley Shear Zone. Gillett and Van Alstine [1982] saw a similar discrepancy in declination between their sites and

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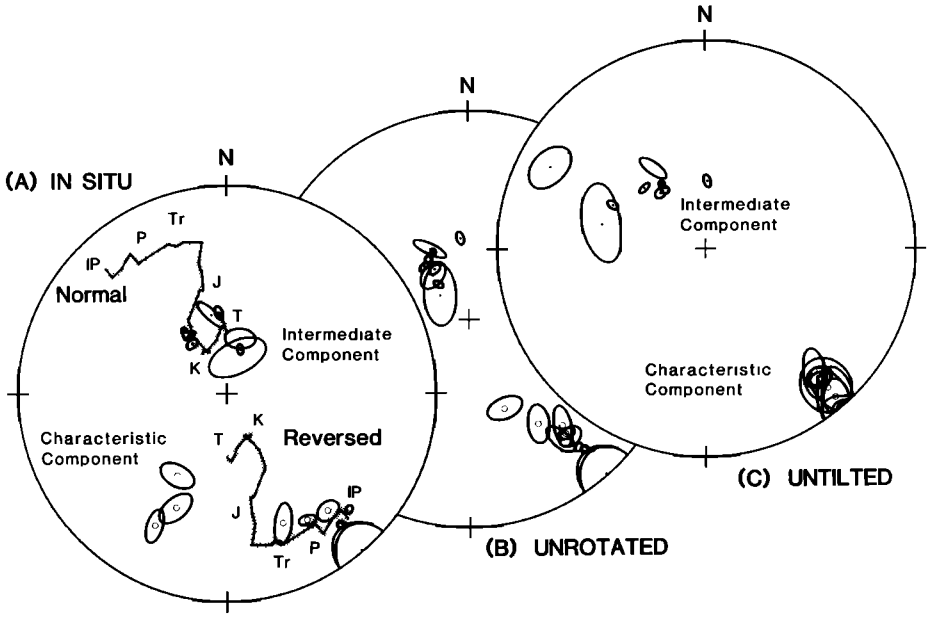


Fig. 5. (a) Equal angle plot showing the predicted magnetic field direction for the Las Vegas Range since the late Paleozoic derived from the mean APW path for North America of Irving and Irving [1982]. Stippled band represents the 3°-4° uncertainty in the APW path. Geologic period abbreviations: ℙ, Pennsylvanian; P, Permian; Tr, Triassic; J, Jurassic; K, Cretaceous; and T, Tertiary. Also plotted are 95% Bingham error ellipses for the components of magnetization measured at the eight sample sites in present-day coordinates. Means plot as dots on the lower hemisphere, open circles on the upper hemisphere. Stars mark the mean tilt-corrected directions for the three northernmost, unrotated sites. (b) Equal angle plot showing the 95% Bingham error ellipses for the intermediate and characteristic magnetization directions after removing the vertical axis rotation due to movement on the LVVSZ. Note the clustering of the intermediate directions around the mean direction for the three northernmost sites (star). This mean direction corresponds to the middle Cretaceous magnetic field direction for the Las Vegas Range. (c) Equal angle plot showing error ellipses for the intermediate and characteristic magnetization directions after correcting for both the vertical axis rotation due to the LVVSZ and tilting due to Sevier thrusting along the Gass Peak Thrust. The intermediate magnetization directions scatter, indicating that the intermediate magnetization post-dates tilting. The characteristic magnetization directions, however, cluster around the late Paleozoic magnetic field direction for the Las Vegas Range, indicating that it predates both the tilting and the vertical axis rotation.

the expected field direction and attributed it to vertical axis rotations related to motion on the LVVSZ south of the Desert Range and the Pahranagat Shear Zone to the north.

The intermediate component of magnetization, as measured at the three least rotated, northernmost sites, NM, CM, and UP (D = 326°, I = 53°, before tilt correction), is within 5° of the late Mesozoic magnetic field direction for the Las Vegas Range (Figure 5a). In contrast to the characteristic component, the intermediate component does not pass the inclination-only fold test at the 95% confidence level (Table 2), indicating that it was imposed either while the beds were being tilted or afterwards. This component appears to have been acquired during uplift associated with movement on the Gass Peak Thrust.

TECTONIC IMPLICATIONS

The paleomagnetic evidence shows that parts of the Las Vegas Range have

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TABLE 1. Summary of Bingham Statistics for Paleomagnetic Sites

Site		Declination	Inclination	K_1	K_2	95% angles		N/No
			Intermediate Component					
NM	in situ	327.7	53.1	-515.73	- 76.55	1.3	3.3	12/13
NM	tilt-corrected	322.3	45.0	-455.77	- 81.88	1.3	3.2	12/13
CM	in situ	325.5	57.4	-119.04	- 42.64	2.0	3.4	21/23
CM	tilt-corrected	320.8	52.0	- 97.07	- 50.10	2.2	3.1	21/23
UP	in situ	326.0	49.7	-349.20	-194.80	1.2	1.6	19/21
UP	tilt-corrected	319.9	48.8	-233.35	-203.88	1.5	1.6	19/21
LP	in situ	349.0	47.7	-137.25	- 12.19	2.5	8.6	12/18
LP	tilt-corrected	344.7	40.6	-114.08	- 16.87	2.7	7.2	12/18
YF	in situ	354.4	47.2	-429.23	-142.14	1.8	3.2	7/21
YF	tilt-corrected	1.8	53.9	-426.20	-142.50	1.8	3.2	7/21
CK	in situ	13.6	59.4	- 24.71	- 10.60	5.1	7.9	17/22
CK	tilt-corrected	350.6	9.9	- 24.01	- 10.01	5.2	8.1	17/22
LO	in situ	16.8	64.8	-317.88	-108.64	1.6	2.8	12/20
LO	tilt-corrected	353.7	37.6	-319.47	-108.73	1.6	2.8	12/20
IR	in situ	15.4	69.5	- 17.81	- 5.40	8.6	16.4	9/13
IR	tilt-corrected	348.9	35.8	- 19.31	- 4.92	8.3	17.4	9/13
			Characteristic Component					
NM	in situ	147.6	-18.5	-253.23	- 77.38	1.8	3.3	12/13
NM	tilt-corrected	146.3	-10.5	-161.56	- 73.26	2.3	3.4	12/13
CM	in situ	139.3	-16.8	- 46.55	- 26.46	3.2	4.3	21/23
CM	tilt-corrected	138.5	-11.2	- 43.30	- 25.92	3.4	4.4	21/23
UP	in situ	133.6	-12.1	-409.27	-222.42	1.1	1.5	19/21
UP	tilt-corrected	132.6	-10.2	-485.36	-197.61	1.0	1.6	19/21
LP	in situ	156.6	-21.7	- 84.88	- 13.99	3.2	8.0	12/18
LP	tilt-corrected	156.4	-13.6	- 61.76	- 12.31	3.8	8.6	12/18
YF	in situ	139.3	0.5	- 25.09	- 18.66	7.8	9.0	7/21
YF	tilt-corrected	139.3	- 8.4	- 25.02	- 18.67	7.8	9.0	7/21
CK	in situ	211.2	-40.7	- 27.43	- 9.41	4.8	8.4	17/22
CK	tilt-corrected	191.6	- 4.0	- 24.33	- 8.62	5.2	8.9	17/22
LO	in situ	204.0	-28.1	- 47.25	- 14.40	4.3	7.9	12/20
LO	tilt-corrected	196.2	- 6.8	- 47.28	- 14.41	4.3	7.9	12/20
IR	in situ	208.7	-18.4	-135.27	- 28.50	2.9	6.3	9/13
IR	tilt-corrected	204.4	3.3	- 77.19	- 38.07	3.8	5.4	9/13

undergone over 60° of relative tectonic rotation around a vertical or near-vertical axis. The declination of the characteristic magnetization increases more than 60° from site UP, 24.4 km north of the axis of the Las Vegas Valley, to site LO, 12.8 km north of it. Likewise, the declination of the intermediate component increases 50.8° ± 4.8° over the same 11.6 km interval (Table 1, Figure 4).

At the northernmost sites (UP, NM, CM) the intermediate component can be very accurately measured and indicates that there has been very little rotation between these three sites since the Sevier orogeny. Hence, we measure all rotations due to the LVVSZ with respect to these relatively fixed sites. In contrast to the intermediate magnetization direction the declination of the characteristic magnetization

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TABLE 2. Inclination-Only Fold Test

Without Structural Correction		With Structural Correction		
Mean Inclination	k_1, Estimate of the Parameter k	Mean Inclination	k_2, Estimate of the Parameter k	k_2/k_1
	For the Intermediate Component			
56.9°	48.76	42.0°	16.91	0.347
	For the Characteristic Component			
- 19.8°	23.35	- 7.7°	116.81	5.00

increases 13.7° ± 3.6° in the 15.5 km from site UP to NM. This is probably due to minor, local faulting and folding which occurred after the acquisition of the characteristic magnetization but before the acquisition of the intermediate magnetization. Similarly, local structures are probably also responsible for the 17.1° ± 9.8° difference in the declination of the characteristic magnetization between sites YF and LP, which are less than 2 km apart. Apparently there are ±10° variations in the declination of the characteristic magnetization due to smaller-scale structures superimposed upon the much larger systematic rotation of both components of magnetization due to movement on the LVVSZ.

Since the characteristic magnetization was acquired, the rocks of the Las Vegas Range have undergone two significant rotations: a tilt around a roughly north-south, horizontal axis due to Sevier age thrusting, followed by a near-vertical axis rotation due to movement on the LVVSZ. In contrast, the intermediate component has only been affected by the vertical axis rotation. To use the characteristic component as a structural marker to determine the amount of vertical axis rotation, the effects of the horizontal axis rotation must be removed using a simple tilt correction. When this is done, the inclinations of the characteristic magnetization are between 3° and -14°, but the declinations range from 133° and 204°. The difference in declination between the three northern, unrotated sites and the other sites reflects the amount of rotation due to movement on the LVVSZ.

When this vertical axis rotation is subtracted from the in situ directions for both the characteristic and intermediate component, both components cluster nicely. The intermediate component directions cluster around a Cretaceous or early Tertiary field direction, while the characteristic magnetization directions move toward a late Paleozoic field direction (Figure 5b). When a tilt correction is applied to restore bedding to horizontal, the intermediate component directions scatter, while the characteristic magnetization directions cluster further (Figure 5c), reaffirming our inference that the characteristic direction predates tilting while the intermediate direction was acquired after tilting.

In contrast, Gillett and Van Alstine [1982] concluded that the intermediate component of magnetization they measured in the Desert Range predated tilting. This is explained by the differing deformational histories of the Las Vegas Range and the Desert Range. In the Desert Range, bedding dips steeply (40-60°) as a result of listric normal faulting during Basin and Range extension in the late Tertiary, well after the intermediate magnetization was acquired. About 100% extension occurred in the Sheep and Desert ranges, while the Las Vegas Range was almost unaffected [Guth, 1980]. The gentler dips of bedding in the Las Vegas Range were acquired as a result of thrusting along the Gass Peak Thrust during the Sevier Orogeny, before the intermediate magnetization was acquired. The Desert Range did not undergo significant tilting during the Sevier orogeny [Guth, 1980].

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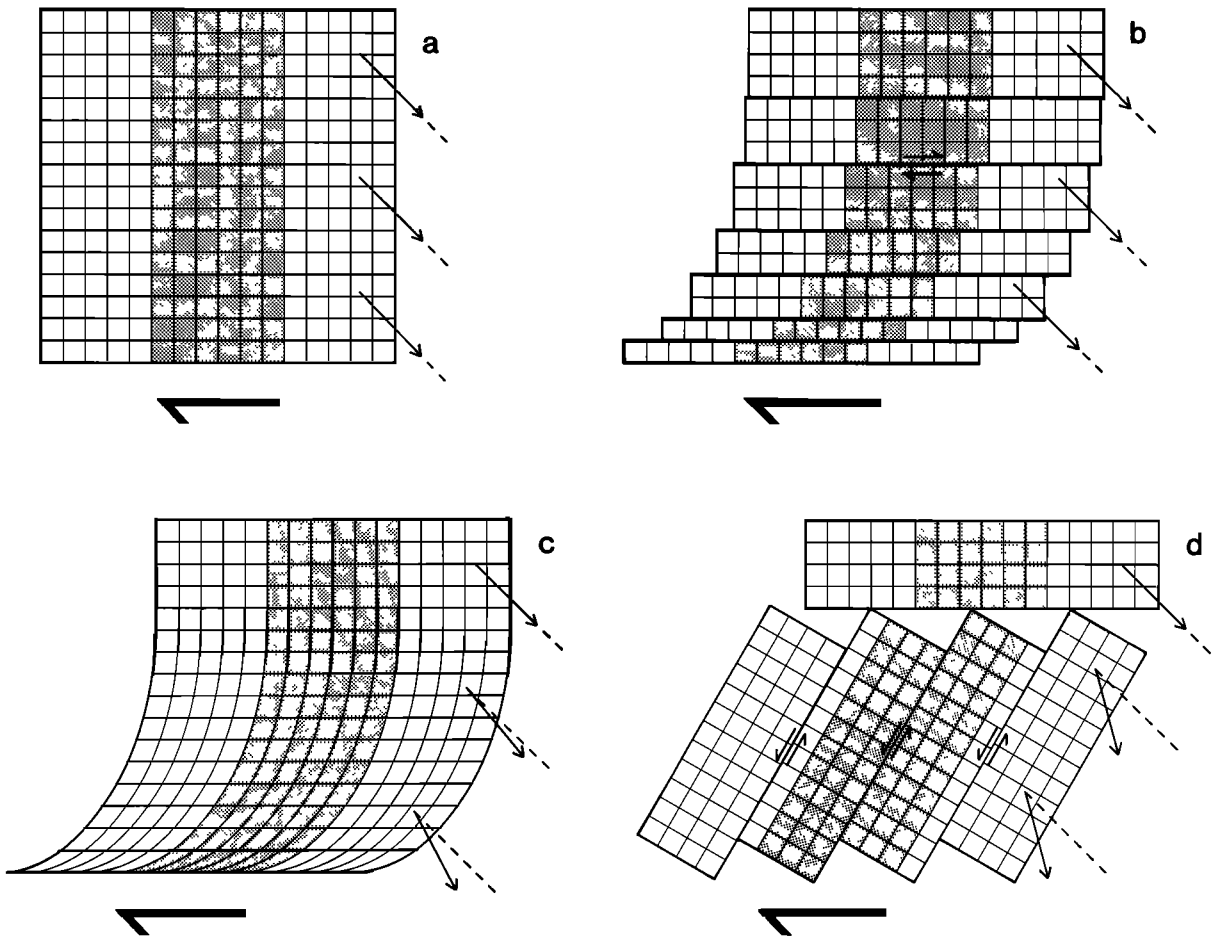


Fig. 6. Possible mechanisms of deformation in a shear zone. (a) An undeformed region prior to development of a shear zone. The stippled zone represents a marker which defines the structural trend of the region. Arrows represent the direction of an ancient magnetization direction. Dashed line shows the unrotated magnetization direction. (b) Shearing along faults parallel to the main shear zone. There is no rotation and hence no change in the direction of the ancient magnetization direction. (c) Pervasive, continuous simple shear. The paleomagnetic samples themselves are deformed and the magnetization direction altered; how much depends upon the mechanism of deformation operating at the microscopic scale. This may be occurring at depth beneath the Las Vegas Range. (d) The large block model of Ron et al. [1984]. Displacement across a shear zone is accommodated by the two bounding strike-slip faults and blocks rotating between them. The amount of rotation inferred from paleomagnetic data is the same everywhere in the shear zone. (e) The small block model. The brittle upper crust is broken into small blocks which rotate independently in response to the continuous, ductile deformation at greater depth. The blocks rotate with an angular velocity equal to half the vorticity of the deforming continuum beneath them. The rigid blocks shown are separated by faults and smaller rigid blocks. Note that the amount of block rotation will be greater than the change in general structural trend. In particular, some blocks may rotate more than 90°, while the change in general structural trend will only approach 90°.

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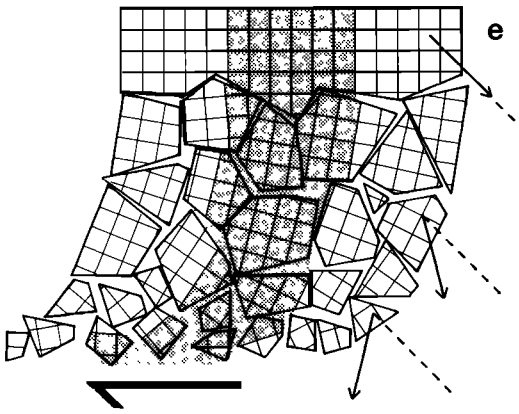


Fig. 6. (continued)

MECHANISMS OF OROFLEXURAL BENDING

Several studies have used paleomagnetism to document tectonic rotations, but few have attempted to explain how these rotations occur. Most workers ascribe the rotations to the movement of rigid blocks without documenting the sizes of the blocks, the nature of their boundaries, or even whether the blocks are indeed rigid. There are a variety of possible mechanisms for deformation in and near shear zones, including those proposed by Ron et al. [1984], McKenzie and Jackson [1983], and England et al. [1985]. We will consider a number of these to see if they apply in the Las Vegas Range.

The first and simplest mechanism accommodates displacement on a shear zone along a series of faults parallel to the main shear zone (Figure 6b) and is similar to that proposed by Geissman et al. [1984] for shear across the Walker Lane in west central Nevada. Such a mechanism could explain the outcrop pattern seen in the Las Vegas Range. However, the blocks between the faults would not be rotated, and thus such a mechanism cannot explain the paleomagnetic data from the Las Vegas Range.

Another possible mechanism for deformation near shear zones is that of pervasive simple shear (Figure 6c). In such a case, shearing would be accommodated by recrystallization and deformation of individual grains in the rock, and the paleomagnetic samples themselves would have been deformed. During Miocene time the rocks of the Las Vegas Range were not deeply buried and were not heated enough to reset the characteristic magnetization, which was acquired when the rocks were at ≈250°C in late Paleozoic time [Gillett and Van Alstine, 1982]. Recrystallization certainly occurs at the pressures and temperatures reached at mid-crustal depths but is unlikely to have affected these rocks. There is no macroscopic textural evidence in the samples for pervasive deformation, but there is abundant field evidence of brittle faulting. In general, the sites sampled were located in apparently rigid blocks well away from zones of faulting and folding.

In a third model, offset across a shear zone is accommodated by slip along mappable, strike-slip, bounding faults and by rotation of rigid blocks within the shear zone (Figure 6d). Ron et al. [1984, 1986] and Luyendyk et al. [1980, 1985] have invoked such a mechanism to explain discordant paleomagnetic declinations in northern Israel, southern Nevada, and southern California. This type of mechanism cannot explain the paleomagnetic data from the Las Vegas Range, because the amount of rotation increases smoothly toward the LVVSZ, indicating that if there are rotating rigid blocks, they do not span the entire shear zone. In addition, this model cannot explain the gradual change in the structural trend from north-south to nearly east-west near the LVVSZ.

McKenzie and Jackson [1983] considered the case where the fault-bounded blocks are much smaller than the width of the deforming zone and rotate in response to continuous deformation in the ductile material beneath them (Figure 6e). They showed that for a rigid block surrounded by faults and deforming zones and floating on a continuum the rotation rate of the block will be $\omega/2$, where ω is the vorticity in the ductile substrate. In such a case the continuous, ductile deformation at middle and lower crustal levels controls the movement of the brittle rock near the surface. In the Las Vegas Range it appears that more than just the upper crust is involved in the deformation, since the rotation of magnetization directions (and hence of

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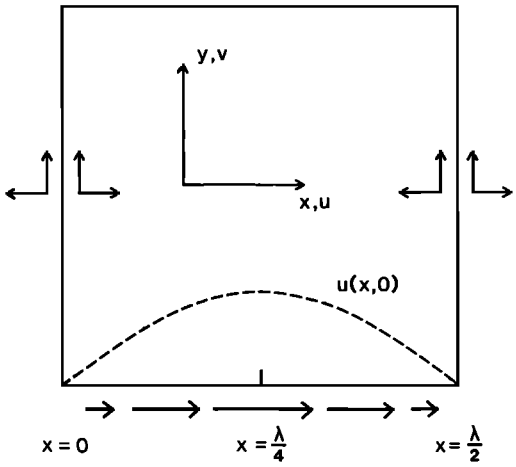


Fig. 7. Summary of the thin viscous plate model of England et al. [1985]. A strike-slip fault follows the x axis. Positive values of velocity component u correspond to left-lateral slip on the fault. The velocity boundary condition shown is imposed upon the fault. Reflecting boundary conditions are imposed on x = 0 and x = λ/2.

the upper crust) up to 20 km from the LVVSZ indicates that a comparable thickness of crust is involved in the deformation.

The paleomagnetic data indicate that this last mechanism is applicable, since the rocks of the Las Vegas Range are broken into a number of small blocks, each one rotated by a different amount. The paleomagnetic declination changes significantly over a range of less than 5 km (for example, between sites LP, YF, and CK). Since the geologic mapping of the Las Vegas Range [Guth, 1980; Ebanks, 1965] does not show any crosscutting faults with enough offset to accommodate the 10°-20° of rotation between sites, it seems likely that the Las Vegas Range is actually broken into many blocks ranging in size from a few kilometers to a few meters in length, separated by small faults and fractures. Such small-scale structures would be too small to appear on any but the most detailed geologic maps and yet can account for much of the deformation in a given area. In a detailed study of a particularly well-exposed part of the Sierra Nevada, Lockwood and Moore [1979] found that microfaults accommodated ≈2.3% extension in an area previously thought to be undeformed due to the lack of large mappable structures. Similarly, Riedel et al. [1984] measured discordant paleomagnetic declinations in the Pomona flow of the Columbia Plateau Basalt Group that indicated rotations which could not be explained by mapped geologic structures. They concluded that movement along much smaller scale fractures and faults was responsible for the rotations.

Thus, because the block size is smaller than the width of the deformed zone, we would expect the amount of rotation to equal half the vorticity ω of the underlying ductile layer. The key question then is to determine how deformation will occur in a continuous ductile solid cut by a strike-slip fault.

THIN VISCOUS PLATE MODEL

England et al. [1985] and Sonder et al. [1986] have done numerical experiments using a thin viscous plate model to examine the length scales of deformation along a strike-slip fault. In contrast to McKenzie and Jackson [1983], who calculated rotations in a shear zone given the total offset and width of the shear zone, Sonder et al. [1986] calculate both the amount of rotation and how it decreases away from the shear zone for a given fault offset, fault length, and plate rheology.

England et al. [1985] and Sonder et al. [1986] consider a thin viscous plate with a fault parallel to the x axis. The components u and v of the velocity are taken to be parallel to the x and y axes, respectively (Figure 7). A tangential velocity boundary condition

$$u(x,0) = U_0 \sin(2\pi x/\lambda) \quad (1)$$

is imposed on the x axis between x = 0 and x = λ/2, where λ/2 is the length of the fault. On the boundaries x = 0 and x = λ/2 a reflecting boundary condition is imposed. England et al. [1985] first assumed a Newtonian viscosity and derived exact solutions for both plane strain deformation and for deformation of a thin viscous sheet in a gravity field normal to the sheet.

For the case of plane strain deformation,

$$u = U_0 \sin(2\pi x/\lambda)\, e^{-2\pi y/\lambda}(1 - 2\pi y/\lambda) \quad (2a)$$

$$v = -U_0 \cos(2\pi x/\lambda)\, e^{-2\pi y/\lambda}(2\pi y/\lambda) \quad (2b)$$

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For the case of a thin viscous sheet,

$$u = U_0 \sin(2\pi x/\lambda)\, e^{-2\pi y/\lambda}[1 - \tfrac{3}{5}(2\pi y/\lambda)] \quad (3a)$$

$$v = -U_0 \cos(2\pi x/\lambda)\, e^{-2\pi y/\lambda}[\tfrac{3}{5}(2\pi y/\lambda)] \quad (3b)$$

Now, the instantaneous angular velocity of a small, horizontal rigid block is

$$\underline{\Omega} = \underline{\omega}/2 = \tfrac{1}{2}\, \underline{\nabla} \times \underline{v}$$

$$= \tfrac{1}{2}(\partial v/\partial x - \partial u/\partial y)\hat{\underline{z}} \quad (4)$$

From this we can calculate the rotations for points near the line $x = \lambda/4$ for comparison with the rotations measured in the Las Vegas Range. For points near $x = \lambda/4$, $u \gg v$ and $\partial u/\partial x$ will be small. Thus for a given point, $\underline{v}$ and $\underline{\Omega}$ will remain almost constant through time even though the point will move with respect to the line $x = \lambda/4$ during deformation. So if t is the duration of deformation, the amount of rotation

$$R \approx |\underline{\Omega}|t = -t/2\ \partial u/\partial y$$

$$= U_0 t \sin(2\pi x/\lambda)\, e^{-2\pi y/\lambda}(2\pi/\lambda)$$

$$\approx U_0 t\, e^{-2\pi y/\lambda}(2\pi/\lambda) \quad (5)$$

for plane strain and

$$R \approx |\underline{\Omega}|t = -t/2\ \partial u/\partial y$$

$$\approx \tfrac{4}{5} U_0 t \sin(2\pi x/\lambda)\, e^{-2\pi y/\lambda}(2\pi/\lambda)$$

$$\approx \tfrac{4}{5} U_0 t\, e^{-2\pi y/\lambda}(2\pi/\lambda) \quad (6)$$

for a viscous plate.

England et al. [1985] noted that these two cases will bound all plausible Newtonian models. Note that the predicted rotations differ by only 25%. For a left-lateral fault the rotations will be counterclockwise; for a right-lateral fault like the LVVSZ the rotations will be clockwise.

We can compare the rotations observed in the Las Vegas Range with these expressions. The Las Vegas Range trends roughly perpendicular to the LVVSZ and is located near the midpoint of the fault, where displacement appears to be greatest. To the west-northwest, displacement decreases until in the Specter Range 55 km to the west there is no surface evidence for displacement [Burchfiel, 1965]. Fifty-five kilometers to the east the LVVSZ is cut off by the Lake Mead Shear Zone [Bohannon, 1979]. So the minimum length of the shear zone is ≈110 km (and thus $\lambda \approx 220$ km).

Just how the displacement varies along the shear zone is not known, but let us assume the simple distribution of slip (Figure 7) used by England et al. [1985]. We do know that the Gass Peak Thrust in the Las Vegas Range has been offset at least 43 km westward to Indian Ridge, so we can make a minimum estimate for the displacement. Because the model is for just one side of the shear zone, we halve the minimum displacement to 22 km. Near the center of the model fault (where $x \approx \lambda/4$) the tangential velocity

$$u(x,0) = U_0 \sin(2\pi x/\lambda)\, e^{-2\pi y/\lambda} \approx U_0 \quad (7)$$

So the total displacement $D \approx U_0 t$, for $D \ll \lambda$.

Comparison With Paleomagnetic Data

In Figure 8a the relative rotation of the characteristic component of magnetization directions is plotted as a function of distance from the Las Vegas Valley Shear Zone. The rotations are taken relative to 139°, which is approximately the average declination measured at the northern three, least deformed sites and is close to the predicted declination for the Las Vegas Range when these rocks were remagnetized in late Paleozoic time. The uncertainty in declination equals half the range in declination of the Bingham 95% error ellipse at each site. These rotations are compared with (6) for a range of values of maximum displacement ($\approx U_0 t$). For plane strain the rotations would be 25% greater than those plotted (compare (5)). For values appropriate for the LVVSZ, England et al.'s Newtonian viscosity models fail to predict rotations of the magnitude documented in the Las Vegas Range. Furthermore, the calculated values decrease more slowly with distance from the shear zone than do the measured differences in declination.

England et al. [1985] also examined numerically the deformation of non-Newtonian viscous plates. For points near the line $x = \lambda/4$ they found that

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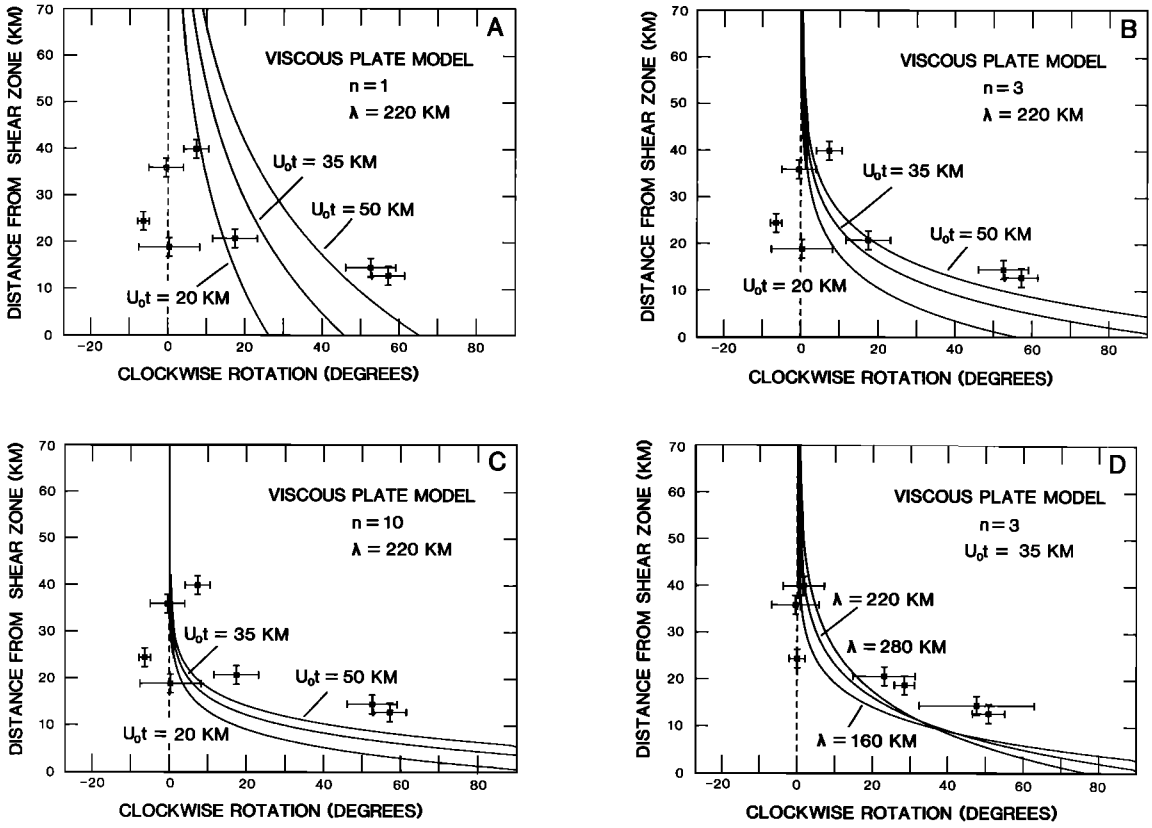


Fig. 8. (a) Plot of the rotation of the characteristic magnetization direction versus distance from the LVVSZ. The uncertainty in amount of rotation equals half the range in declination of the 95% Bingham error ellipse. The 2-km uncertainty in distance from the shear zone is due to the alluvium of the Las Vegas Valley, which conceals the shear zone. The curves show the amount of rotation predicted by the viscous plate model of England et al. [1985] for n = 1 (i.e., a Newtonian rheology) and λ = 220 km with three values of U_0t ($\approx$ D, the maximum displacement on the fault). (b) Same as Fig. 8a with n = 3. (c) Same as Fig. 8a with n = 10. (d) Plot of the rotation of the intermediate magnetization direction versus distance from the LVVSZ. The curves show the predicted rotations for a viscous plate model for n = 3 and U_0t = 35 km with three values of λ (= twice the fault length).

$$u \approx U_0 \sin(2\pi x/\lambda)\, e^{-4\sqrt{n}\pi y/\lambda} \qquad (8)$$

where n is the stress exponent.

Thus, since $\partial v/\partial x \ll \partial u/\partial y$ for $x \approx \lambda/4$,

$$R \approx -t/2\ \partial u/\partial y$$

$$\approx (2U_0 t\pi\sqrt{n}/\lambda) \sin(2\pi x/\lambda)\, e^{-4\sqrt{n}\pi y/\lambda} \qquad (9)$$

This expression is plotted in Figures 8b and 8c for different values of the U_0t ($\approx$ displacement) for both n = 3 and n = 10. In addition, the amount of rotation inferred from the declination of the characteristic magnetization is plotted. The amount of rotation inferred from the declination of the intermediate magnetization is plotted in Figure 8d along with model curves for different values of λ to show how the amount of rotation predicted varies with fault length (= $\lambda/2$). It is clear that much better agreement is obtained for n = 3 to 10 than for n = 1. Although the calculated curves do not exactly match the actual data, they do predict both the maximum amount of rotation and the length scale of deformation, especially for the larger values of offset U_0t.

There are a number of possible reasons

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why the paleomagnetic data do not match the theoretical curves more closely. The most important is that the crustal "blocks" sampled will not perfectly reflect the continuous deformation occurring at depth (see Figure 6e). The blocks have a finite extent and cannot perfectly mirror the continuous changes in the vorticity of the continuum beneath them.

At some sites (e.g., NM and UP) there is a large (≥10°) discrepancy between the amount of rotation inferred from the intermediate component of magnetization and the characteristic component (Table 3). As discussed above, this could be due to small rotations that occurred between the acquisition of the two components. In addition, the error in isolating an intermediate component of magnetization is greater than that in determining the direction of a characteristic magnetization. While the characteristic component of magnetization is more easily isolated, it has been affected by more tectonic events than the younger intermediate component. Another possible explanation for the discrepancy between the two components is that movement along the LVVSZ caused rotations around slightly nonvertical axes. Because the intermediate direction plunges steeply, a small change (≈5°) in the axis of rotation would significantly alter the change in declination for a given amount of rotation. In contrast, the nearly horizontal characteristic component is much less sensitive to such a perturbation of the axis of rotation. The discrepancy in the amount of rotation inferred from the two components of magnetization is greater than the uncertainty in the declination of the individual components (ΔD in Table 1) and provides a conservative estimate of the uncertainty in the inferred rotation.

Comparison With Structural Data

The near-horizontal characteristic magnetization direction provides an invaluable marker for determining the amount of rotation in the Las Vegas Range because geologic mapping alone cannot definitively determine the amount of rotation. The amounts of clockwise rotation of the characteristic component of magnetization do not equal the differences in the strike of bedding between sample sites (Table 3). This is particularly clear at the three northern sites where the beds are subhorizontal. At most sites the strike varied up to 20° because a small change in the orientation of a subhorizontal bed can cause a large change in strike. Thus the strike of bedding does not provide an adequate marker for determining vertical axis rotations in the Las Vegas Range.

However, other structural data can be used to test the deformation mechanism inferred from paleomagnetic data in the Las Vegas Range. Geologic mapping by Longwell [1965], Guth [1980], and Ebanks [1965] showed that the Gass Peak Thrust, the associated folds which parallel it, and the contacts between units all trend nearly north-south in the northern Las Vegas Range but become increasingly east-west closer to the LVVSZ (Figure 4). These structures were formed by east-directed thrust faulting during the Sevier orogeny and consistently trend approximately N 10° E in southern Nevada. Everywhere the Gass Peak Thrust is exposed in the Las Vegas Range it places Cambrian Stirling Quartzite over Permo-Pennsylvanian Bird Spring Formation. Since the thrust is ramping upsection in the Las Vegas Range, this consistent juxtaposition indicates that erosion has exposed the same structural level everywhere. Thus the trend of the Gass Peak Thrust defines the structural trend particularly well. If we assume that the Gass Peak Thrust ramp originally trended N 10° E, paralleling other Sevier age thrusts well removed from the LVVSZ, we can use the trace of the fault to determine the change in structural trend due to the Miocene motion on the LVVSZ.

The structural trend, as defined by the trace of the Gass Peak Thrust, initially ran nearly perpendicular to the LVVSZ. The general trend of a passive marker initially perpendicular to a shear zone where $v = 0$ will rotate with an angular velocity of magnitude

$$\dot{\theta} = \tan^{-1}(\partial u/\partial y) \qquad (10)$$

(For a more general expression, refer to McKenzie and Jackson [1983, equation 18].) Using the non-Newtonian viscous plate model of England et al. [1985] we can find the rotation of this marker:

$$\theta \approx \dot{\theta} t$$

$$\approx \tan^{-1}[(4U_0 t\pi\sqrt{n}/\lambda)e^{-4\sqrt{n}\pi y/\lambda}$$

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TABLE 3a. Amount of Relative Rotation of the Characteristic Component (Tilt-Corrected)

Site	Distance From the Axis of the Las Vegas Valley	D, Declination of the Characteristic Component	Uncertainty ΔD	Rotation Relative to 139°	Local Attitude of Bedding	Structural Trend
NM	39.9 km	146.3°	3.3°	7.3°	N 26 E 10 W	N 05 E
CM	35.9 km	138.5°	4.4°	-0.5°	N 21 E 7 W	N 05 E
UP	24.4 km	132.6°	1.5°	-6.4°	N 26 W 5 W	N 10 E
LP	20.7 km	156.4°	5.8°	17.4°	N 66 E 9 N	N 25 E
YF	18.8 km	139.3°	7.9°	0.3°	N 46 E 9 S	N 25 E
CK	14.5 km	191.6°	6.4°	52.6°	N 60 E 56 N	N 60 E
LO	12.8 km	196.2°	4.3°	57.2°	N 61 E 32 N	N 70 E
IR	10.6 km	204.4°	4.3°	65.4°	N 61 E 32 N	N 50 E

TABLE 3b. Amount of Relative Rotation of the Intermediate Component (in Situ)

Site	Distance From the Axis of the Las Vegas Valley	D, Declination of the Intermediate Component	Uncertainty ΔD	Rotation Relative to 326°	Local Attitude of Bedding	Structural Trend
NM	39.9 km	327.7°	5.5°	1.7°	N 26 E 10 W	N 05 E
CM	35.9 km	325.5°	6.3°	-0.5°	N 21 E 7 W	N 05 E
UP	24.4 km	326.0°	2.1°	0	N 26 W 5 W	N 10 E
LP	20.7 km	349.0°	8.3°	23.0°	N 66 E 9 N	N 25 E
YF	18.8 km	354.4°	2.7°	28.4°	N 46 E 9 S	N 25 E
CK	14.5 km	13.6°	15.3°	47.6°	N 60 E 56 N	N 60 E
LO	12.8 km	16.8°	4.3°	50.8°	N 61 E 32 N	N 70 E
IR	10.6 km	15.4°	41.1°	49.4°	N 61 E 32 N	N 50 E

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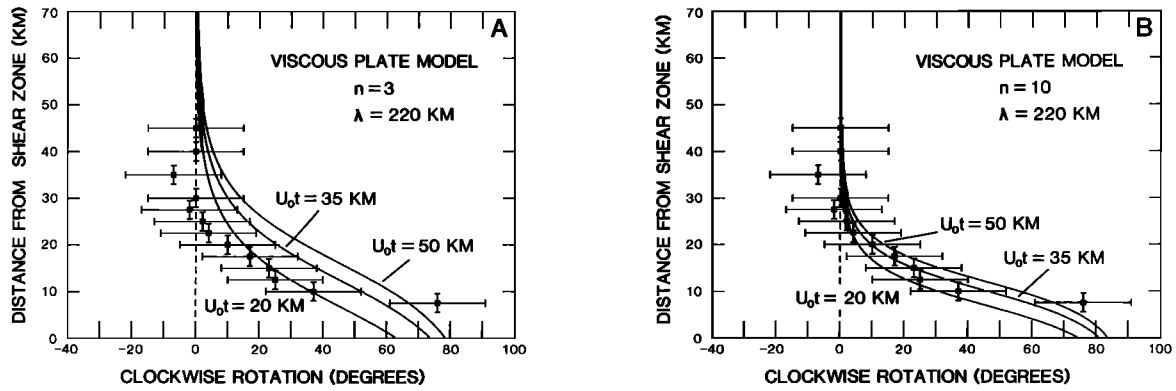


Fig. 9. (a) Plot of the inferred rotation of the structural trend as defined by the Gass Peak Thrust and associated folds versus distance from the LVVSZ. The curves show the change in orientation of a marker originally perpendicular to the LVVSZ predicted by a viscous plate model with n = 3 and λ = 220 km with three values of U_0t. (b) Same as Fig. 9a with n = 10.

$$\cdot \sin(2\pi x/\lambda)] \qquad (11)$$

$$\approx \tan^{-1}[(4U_0t\pi\sqrt{n}/\lambda)e^{-4\sqrt{n}\pi y/\lambda}] \qquad (12)$$

This result is plotted (Figure 9) with the inferred rotation of the trace of the Gass Peak Thrust in the Las Vegas Range as a function of distance from the LVVSZ. Because the Gass Peak Thrust was surely not perfectly linear and because it bifurcates and varies locally in orientation, making it difficult to accurately determine the general trend of structure, this estimate was assigned a rather large uncertainty (±15°). For appropriate values of fault length and displacement, England et al.'s [1985] expression predicts both the observed maximum rotation and the observed decrease in rotation with distance away from the shear zone.

SUMMARY

Thermal demagnetization of samples of the Cambrian Bonanza King Formation in the Las Vegas Range has revealed two components of magnetization: a steeply-dipping, northerly intermediate direction due to a late Mesozoic remagnetization and a more stable, shallow, south to southeasterly direction due to a late Paleozoic remagnetization. We have used these magnetization directions as structural markers to conclude that the rocks of the Las Vegas Range were tilted during Sevier age thrusting, prior to the acquisition of the intermediate component, and then rotated around a vertical axis due to Miocene movement on the LVVSZ. The amount of rotation systematically increases from 0° at sites more than 24 km from the LVVSZ to >50° at sites 10-13 km from the shear zone. By using both ancient components of magnetization we have documented both the amount of rotation between sites and the near-vertical orientation of the rotation axis.

Rotations in the Las Vegas Range due to displacement on the LVVSZ can be explained by the independent rotations of small (<5 km in length) rigid blocks rotating in response to the deformation of a viscous continuum below [McKenzie and Jackson, 1983; Sonder et al., 1986]. Despite the uncertainties of the paleomagnetic data and the simplicity of the models, our results do support the use of non-Newtonian viscous plate models to describe large-scale deformation of the crust. One such model has been used successfully to explain rotations associated with deformation along the >1000-km-long San Andreas Fault system [Sonder et al., 1986]. Our results from the Las Vegas Range indicate that this model is also applicable at smaller length scales (≈100 km), like that of the LVVSZ.

How small (<1 km) faults and fractures accommodate distributed shear has not been considered in many paleomagnetic studies of rotated terranes. Several workers [e.g., Luyendyk et al., 1980; Ron et al., 1984] have used rigid block models to explain their paleomagnetic

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results. Evidence for rotation of paleomagnetic directions from one site is used to infer that all of a large fault-bounded block containing the site has been similarly rotated. Such an assumption cannot be applied in the Las Vegas Range because it has not behaved as a single rigid block. The observed rotations in the Las Vegas Range are due to deformation up to tens of kilometers away from the mappable faults that bound it. Rigid block rotations have occurred, but the blocks must be much smaller than those often postulated. The complexity of such deformation could easily go unrecognized in an area where the outcrop is less continuous or the geologic structure less well known.

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